CSE7101- Capstone Project
Review-3

PSCS_35: 360-Degree Feedback Software for News Stories in Regional Media (using AI/ML)

Batch Number: CSE_150

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Abstract

This project introduces an AI/ML-driven system to analyze regional news stories about government policies and initiatives. The software identifies sentiment (positive, neutral, negative) and classifies articles into relevant departments using NLP techniques. An interactive dashboard presents the results, giving policymakers, media houses, and researchers a comprehensive 360-degree view of public opinion.

- Automates sentiment analysis, reducing manual bias.
- Provides quick and scalable insights from regional news.
- Facilitates better decision-making and policy communication.
- Lays the foundation for future multilingual and real-time analysis.

IEEE Papers Literature Survey

Paper Title	Year	What they did	Why it matters
Deciphering Political Entity Sentiment in News with LLMs	2024	Used LLMs to detect sentiment toward political entities in news.	Shows AI can analyze government-related news sentiment.
Prediction of Stock Value Changes Using Sentiment	2024	Linked news sentiment with stock market changes.	Proves sentiment analysis impacts decision-making.
Fake News Stance Detection Using Deep Learning	2023	Classified stance (support/oppose/neutral) in fake news.	Adds credibility/stance analysis to sentiment work.
Deep Sentiment Analysis: Hausa Case Study	2024	Applied sentiment analysis in a low-resource language.	Guides extension to Indian regional languages.
Multilingual Sentiment Analysis	2024	Reviewed multilingual sentiment approaches (ML/DL).	Informs multilingual scalability in your project.
Enhancing Multilingual Hate Speech Detection	2024	Built NLP models for hate speech in multiple languages.	Helps adapt classification to regional contexts.
Sentiment Analysis of Tweets (Emojis + Texts)	2024	Combined emoji + text for better sentiment accuracy.	Suggests using multi-modal cues beyond text.
Financial Sentiment Analysis with LLMs & FinBERT	2024	Applied domain-specific models for finance news.	Shows specialized models improve accuracy.
Stock Prediction using Headlines Sentiment	2024	Used headlines to predict stock trends with DL.	Justifies using short regional headlines in analysis.
Ensemble Learning for Fake News Detection	2024	Used hybrid multi-modal ensemble for fake news.	Inspires robust ensemble methods for future.

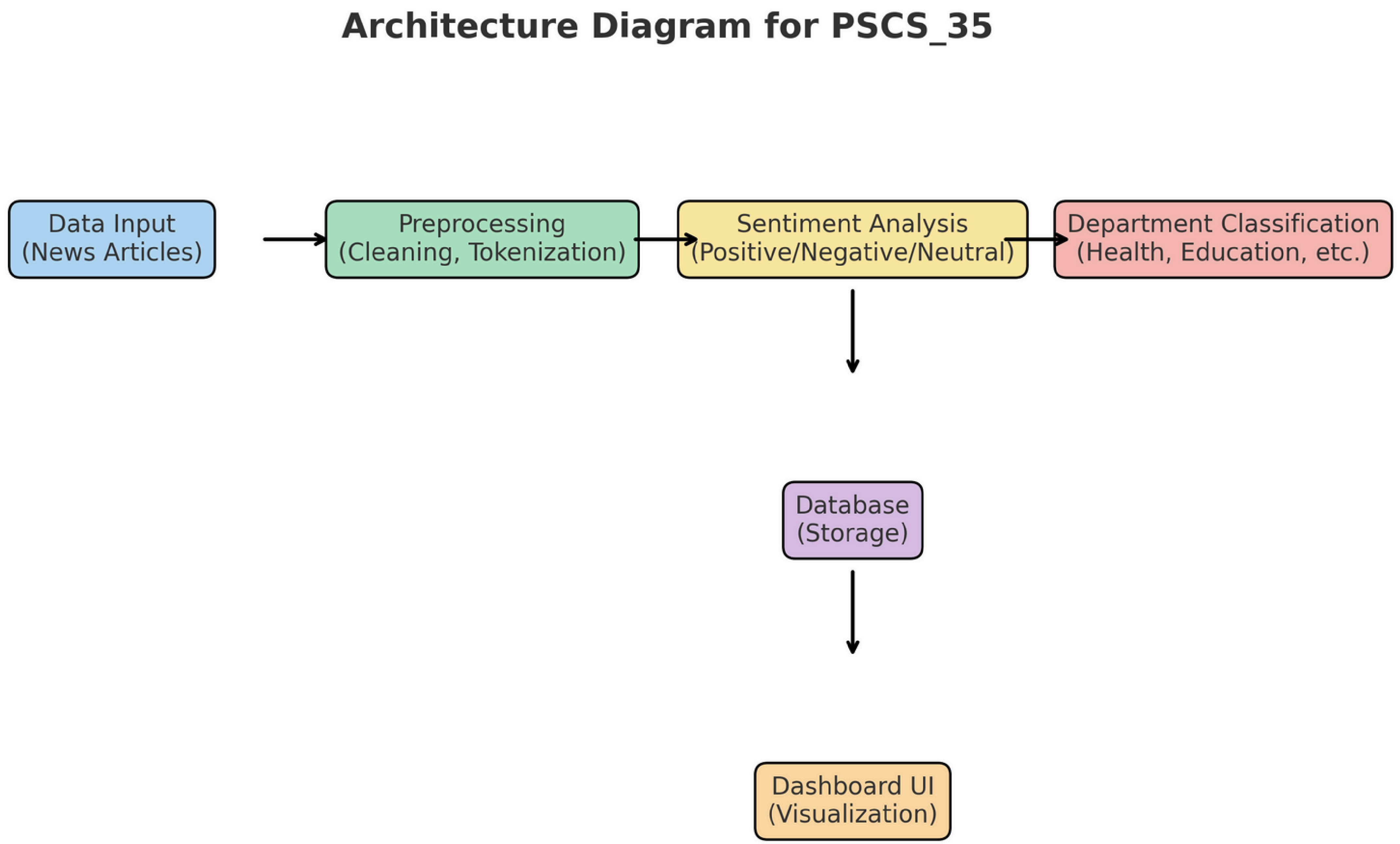
Objectives

Main Goal: Develop an AI/ML-based system for analyzing regional news sentiment and departmental classification.

Specific Objectives:

- Automate sentiment analysis of news (Positive, Negative, Neutral).
- Enable automatic departmental classification of news content (Health, Education, Defense, etc.).
- Build a user-friendly dashboard to visualize sentiment trends and department mapping.
- Provide timely, unbiased feedback to policymakers and media houses.
- Lay foundation for future multilingual support and real-time analysis.

Architecture Diagram



Modules

1. Data Collection Module

- Input from CSV, scraping, or manual entry.

2. Preprocessing Module

- Cleaning, tokenization, stop-word removal.

3. Sentiment Analysis Module

- NLP model determines polarity (Positive/Negative/Neutral).

4. Department Classification Module

- Keyword + ML rules for mapping to departments.

5. Dashboard & Visualization Module

- Flask web app + Chart.js graphs for results.

6. Storage Module (Optional)

- SQLite for persistence & historical analytics.

Hardware and Software Requirements

Hardware (Mac Mini or equivalent):

- Processor: Intel i3 / Apple M1 or higher
- RAM: 8 GB minimum
- Storage: 50 GB free
- Internet: Required for scraping & deployment

Software:

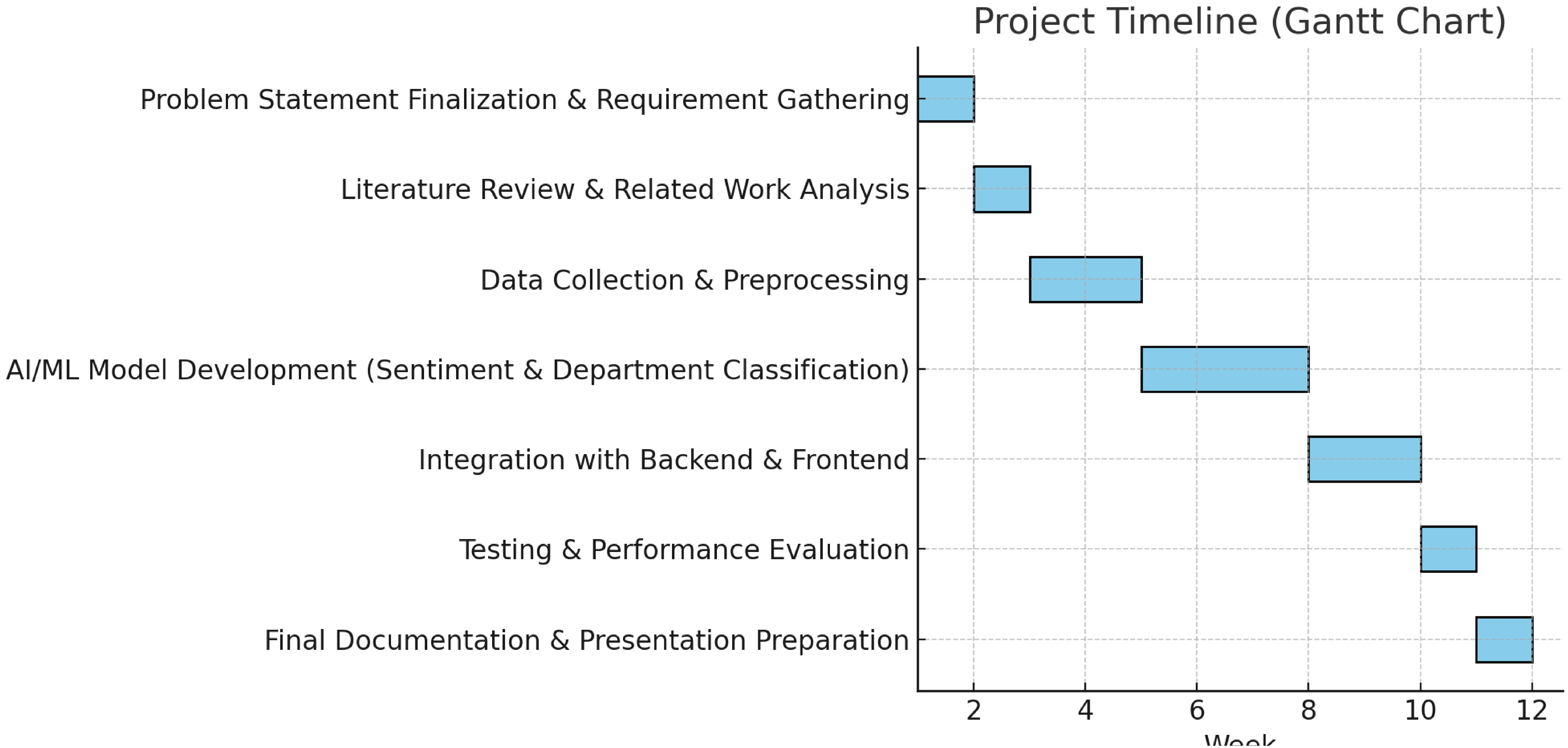
- OS: macOS / Linux / Windows
- Language: Python 3.x
- Libraries: Flask, TextBlob, Pandas, BeautifulSoup4, Chart.js
- Database: SQLite (lightweight, built-in)
- IDE/Editor: VS Code, PyCharm, or Replit online

Github Link

Github Link

https://github.com/YashasRaj20231CSE3019/capstone-project-CSE_150.git

Timeline of the Project (Gantt Chart)



Timeline of the Project (Gantt Chart)

Phases	Task	Duration	Dates
Phase 1	Problem Statement Finalization & Requirement Gathering	1 week	Week 1
Phase 2	Literature Review & Related Work Analysis	1 week	Week 2
Phase 3	Data Collection & Preprocessing	2 weeks	Weeks 3–4
Phase 4	AI/ML Model Development (Sentiment & Department Classification)	3 weeks	Weeks 5–7
Phase 5	Integration with Backend & Frontend	2 weeks	Weeks 8–9
Phase 6	Testing & Performance Evaluation	1 week	Week 10
Phase 7	Final Documentation & Presentation Preparation	1 week	Week 11

References (IEEE Paper format)

All sources referenced in this presentation adhere strictly to the IEEE citation style.

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Thank
You!